

CS 305 Lab Tutorial

Lab10 IP & Multicast

Dept. Computer Science and Engineering
Southern University of Science and Technology

Topic

- IPv4
 - IPv4
 - IP datagram, IP address
 - Multicast
 - Multicast IPv4 address & Mac address
 - Multicast source, Multicast group member, Router
- IPv6
 - IPv6
 - address, header & extension header
 - Multicast
 - Multicast IPv6 address & Mac address
 - Multicast source, Multicast group member, Router

Tips on eNSP : useful commands while in CLI

➤ display

- dis this
- dis ip routing-table
- **dis multicast forwarding-table**
- **dis multicast ipv6 forwarding-table**

➤ enable IPv6

- [Huawei] **ipv6**
- [interface type id-list] **ipv6 enable**

➤ configure interface

- **ip address** [IPv4-address in dotted decimal] [mask]
- **ip address** [IPv6-address] [mask]

➤ configure static routing entry

- ip route-static [destination-netID] [mask] [next-hop]
- **ipv6 route-static** [destination network] [mask] [next-hop]

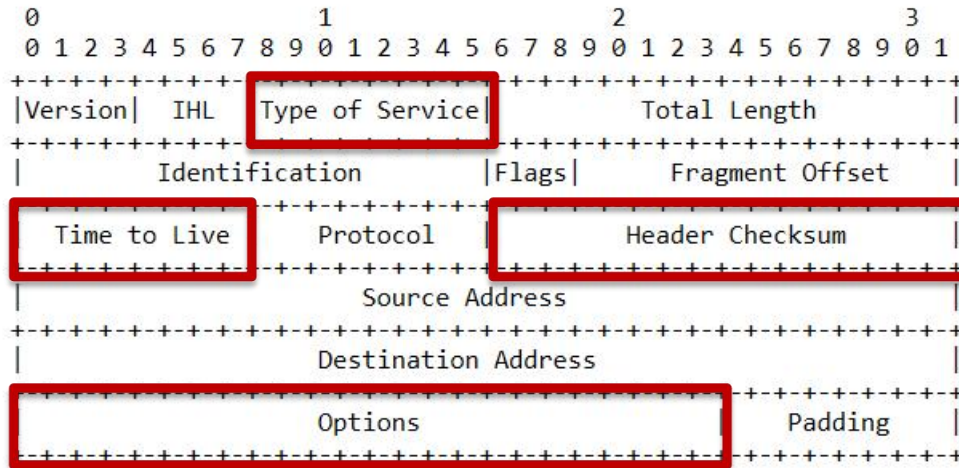
➤ configure multicast

- [Huawei] **multicast routing-enable**
- [interface type id-list] **igmp enable**
- [interface type id-list] **pim dm**
- [interface type id-list] **mld enable**
- [interface type id-list] **pim v6 enable**

Part A. IPv4

- **Best effort** : NO connection, NO flow control, NO congestion control, NO retransmission...
- The internet protocol implements two basic functions: **addressing** and **fragmentation**.
 - The internet modules use the **addresses** carried in the internet header to transmit internet datagrams toward their destinations. The selection of a path for transmission is called **routing**.
 - The internet modules use fields in the internet header to **fragment** and **reassemble** internet datagrams when necessary for transmission through "small packet" networks. The model of operation is that an internet module resides in each host engaged in internet communication and in each gateway that interconnects networks.

IPv4 Datagram



Example Internet Datagram Header

- **Type of Service:**

The major choice is a three way tradeoff between low-delay, high-reliability, and high-throughput.

- **Time to Live (TTL):**

an indication of an upper bound on the lifetime of an internet datagram. **It is set by the sender of the datagram and reduced at the points along the route where it is processed.** An IP datagram with zero TTL will be dropped.

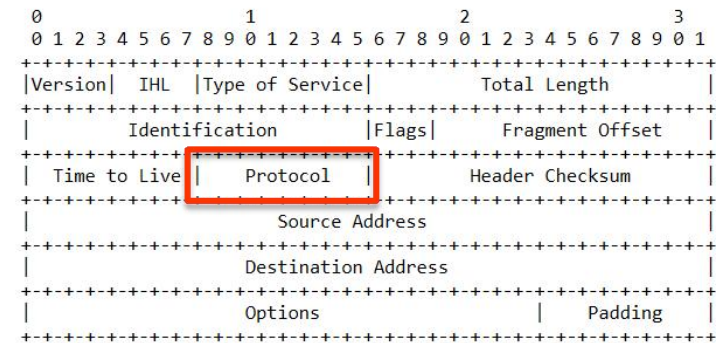
- **Header Checksum:**

provides a verification that the information used in processing internet datagram has been transmitted correctly. The data may contain errors. If the header checksum fails, the internet datagram is discarded at once by the entity which detects the error.

- **Options:**

provide for control functions needed or useful in some situations but unnecessary for the most common communications. The options include provisions for timestamps, security, and special routing.

Protocol Field



Example Internet Datagram Header

- ▼ Internet Protocol Version 4, Src: 192.168.2.104 (192.168.2.104), Dst: t
 - 0100 = Version: 4
 - 0101 = Header Length: 20 bytes (5)
 - > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 - Total Length: 52
 - Identification: 0x05ec (1516)
 - > Flags: 0x4000, Don't fragment
 - Time to live: 64
 - Protocol: TCP (6)
 - Header checksum: 0x0fda [validation disabled]
 - [Header checksum status: Unverified]
 - Source: 192.168.2.104 (192.168.2.104)
 - Destination: tg-in-f113.1e100.net (192.168.2.104)
 - > Transmission Control Protocol, Src Port: 49152, Dst Port: 80

- ▼ Internet Protocol Version 4, Src: tw.net-east.com (116.77.76.254), Dst: 192.168.2.104
 - 0100 = Version: 4
 - 0101 = Header Length: 20 bytes (5)
 - > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 - Total Length: 128
 - Identification: 0x311d (12573)
 - > Flags: 0x0000
 - Time to live: 57
 - Protocol: UDP (17)
 - Header checksum: 0xcbf4 [validation disabled]
 - [Header checksum status: Unverified]
 - Source: tw.net-east.com (116.77.76.254)
 - Destination: 192.168.2.104 (192.168.2.104)
 - > User Datagram Protocol, Src Port: 54321, Dst Port: 80
 - > Domain Name System (response)

- ▼ Internet Protocol Version 4, Src: 192.168.2.104 (192.168.2.104), Dst: 116.7.234.3
 - 0100 = Version: 4
 - 0101 = Header Length: 20 bytes (5)
 - > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 - Total Length: 1020
 - Identification: 0x0a9a (2714)
 - > Flags: 0x00b9
 - Time to live: 6
 - Protocol: ICMP (1)
 - Header checksum: 0x8493 [validation disabled]
 - [Header checksum status: Unverified]
 - Source: 192.168.2.104 (192.168.2.104)
 - Destination: 116.7.234.3 (116.7.234.3)
 - > [2 IPv4 Fragments (2480 bytes): #1(1480), #2(1000)]
 - ▼ Internet Control Message Protocol

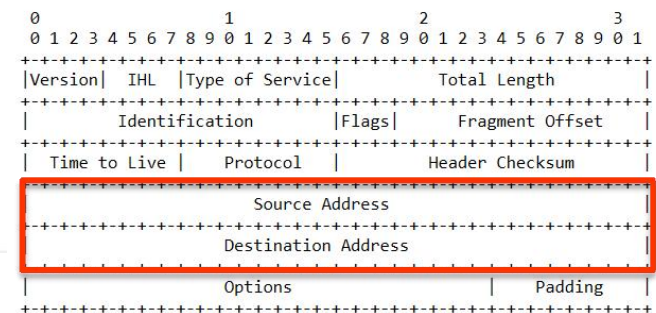
Source and Destination Field

```
> Frame 4: 216 bytes on wire (1728 bits), 216 bytes captured (1728 bits) on interface 0
> Ethernet II, Src: IntelCor_5c:69:58 (90:61:ae:5c:69:58), Dst: IPv4mcast_7f:ff:fa (01:00:5e:7f:ff:fa)
√ Internet Protocol Version 4, Src: 192.168.2.104 (192.168.2.104), Dst: 239.255.255.250 (239.255.255.250)
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 202
    Identification: 0x7437 (29751)
> Flags: 0x0000
    Time to live: 1
    Protocol: UDP (17)
    Header checksum: 0x91e1 [validation disabled]
    [Header checksum status: Unverified]
    Source: 192.168.2.104 (192.168.2.104)
    Destination: 239.255.255.250 (239.255.255.250)
> User Datagram Protocol, Src Port: 58806 (58806), Dst Port: ssdp (1900)
> Simple Service Discovery Protocol
```

```

Internet Protocol Version 4, Src: 0.0.0.0 (0.0.0.0), Dst: 255.255.255.255 (255.255.255.255)
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 328
    Identification: 0xb310 (45840)
  > Flags: 0x0000
    Time to live: 128
    Protocol: UDP (17)
    Header checksum: 0x8695 [validation disabled]
    [Header checksum status: Unverified]
    Source: 0.0.0.0 (0.0.0.0)
    Destination: 255.255.255.255 (255.255.255.255)

```



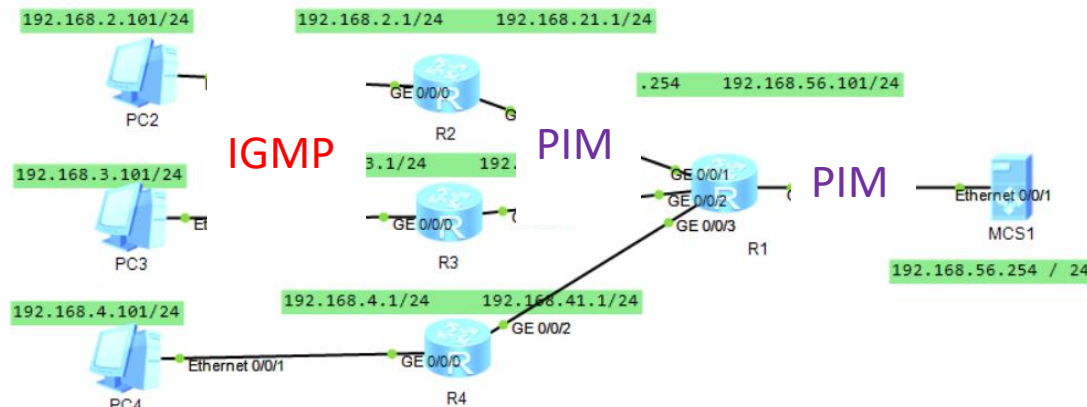
Example Internet Datagram Header

IPv4 Address Categories

- **Unicast , Broadcast , Multicast**
 - **Unicast Address:** A unicast address uniquely identifies a single network interface. Data packets sent to a unicast address are delivered to exactly one specific destination host.
 - **unicast** addresses range from **0.0.0.0 to 223.255.255.255.**
 - **Broadcast Address:** A broadcast address is used to send data packets to all hosts within the same network segment. When a host receives a packet with this address, it processes the packet.
 - **broadcast** address is **255.255.255.255.**
 - **Multicast Address:** A multicast address identifies a group of hosts (not a single host or all hosts). Data packets sent to a multicast address are delivered to all members of the group.
 - **multicast** addresses range from **224.0.0.0 to 239.255.255.255.**

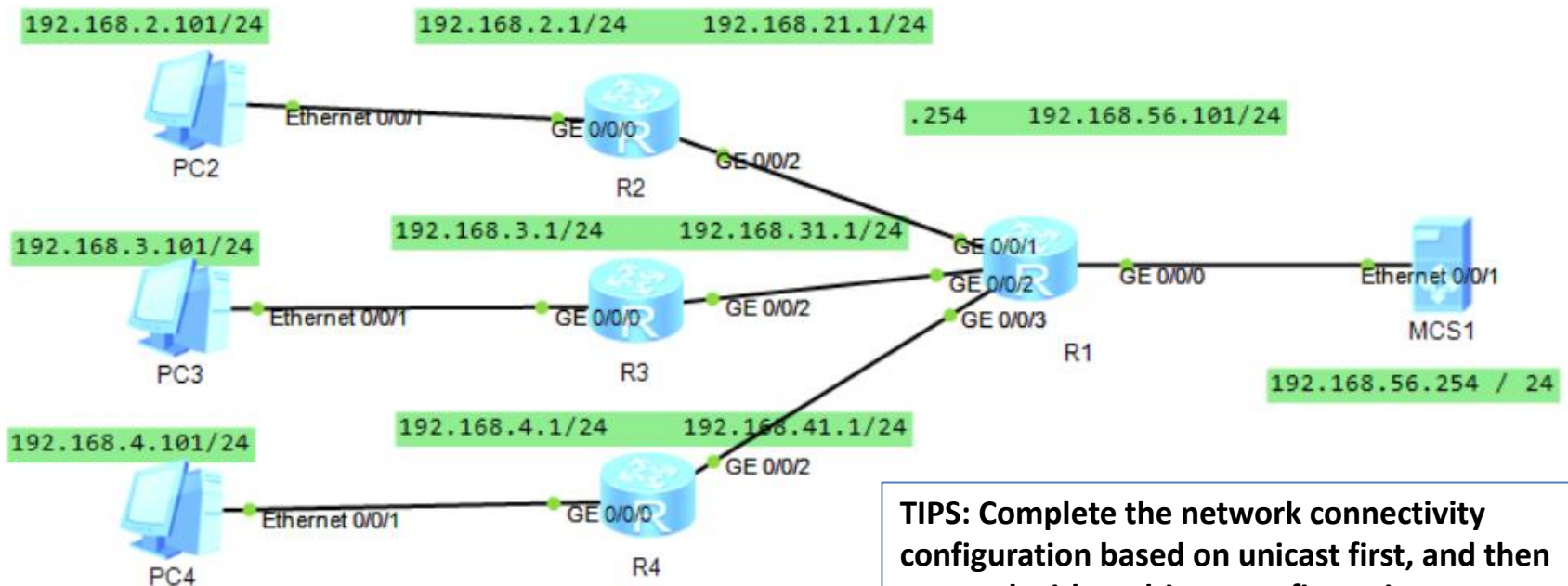
IPv4 Multicast

- **Multicasting** is a network communication method that efficiently **delivers data from one source to multiple recipients**, offering **significant bandwidth savings** compared to unicast and broadcast transmissions.
 - It is particularly valuable for **multimedia applications**(like video conferencing, live streaming, and online gaming) and **collaborative workspaces**.
- **Relevant IPv4 Multicast Protocols**
 - **IGMP (Internet Group Management Protocol)**: Manages multicast group memberships on IPv4 networks by allowing **hosts to join/leave groups** and **informing routers** of their participation.
 - **PIM (Protocol Independent Multicast)**: **A routing protocol** that builds efficient distribution trees for multicast traffic, working alongside existing unicast routing protocols.
 - DM: Best for dense networks
 - SM: Designed for sparse networks



Demo1 (1) - topology

- In this network topology:
 - MCS1 is the **multicast source node**
 - PC2, PC3, PC4 are **potential multicast group member nodes**
 - Routers R1, R2, R3, and R4 are **routers** between multicast source nodes and multicast group nodes, **generating multicast related routes through PIM-DM protocol interaction**
 - R2, R3, and R4 are routers near multicast group member nodes, **sensing changes in multicast groups through IGMP protocol interaction.**



TIPS: Complete the network connectivity configuration based on unicast first, and then proceed with multicast configuration.

Demo1 (2) - multicast group member



PC

PC is **multicast group member nodes** in IPv4 Multicast, Do the configuration on “Basic Config” and “MCPacket”. While choose “239.1.1.1” as the multicast IP address, it’s MAC address is auto-generated as “01-00-5e-01-01-01”.

PC2

Basic Config | Command | MCPacket | UdpPacket | Console

Host Name:

MAC Address: 54-89-98-21-2F-A9

IPv4 Configuration

☒ Static ☐ DHCP ☐ Obtain DNS server address automatically

IP Address: 192 . 168 . 2 . 101 DNS1: 0 . 0 . 0 . 0

Subnet Mask: 255 . 255 . 255 . 0 DNS2: 0 . 0 . 0 . 0

Gateway: 192 . 168 . 2 . 1

MAC Address(48bits) Mapping:

- IPv4 multicast addresses are mapped to MAC addresses using a fixed prefix 01:00:5E.
- The last 23 bits of the IPv4 multicast address are directly mapped to the last 23 bits of the MAC address.

PC2

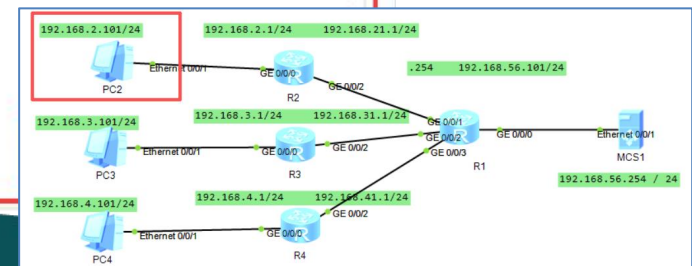
Basic Config | Command | MCPacket | UdpPacket | Console

Packet Configuration

IGMP Version: ☐ Version 1 ☒ Version 2 ☐ Version 3

Source IP: 192 . 168 . 2 . 101 Source MAC: 54-89-98-21-2F-A9

Destination IP: 239 . 1 . 1 . 1 Destination MAC: 01-00-5E-01-01-01



Demo1 (3) - multicast source



MCS

MCS is MultiCast Source nodes in IPv4 Multicast, Do the configuration on “**Basic Config**” and “**MCS**Source”. While choose “**239.1.1.1**” as the multicast IP address, it’s MAC address is auto-generated as “**01-00-5e-01-01-01**”.

MAC Address(48bits) Mapping:

- IPv4 multicast addresses are mapped to MAC addresses using a fixed prefix 01:00:5E.
- The last 23 bits of the IPv4 multicast address are directly mapped to the last 23 bits of the MAC address.

MCS1

Basic Config | MCSSource

Host Name:

Configuration

IP Address:

Subnet Mask:

Gateway:

MAC Address:

MCS1

Basic Config | MCSSource

Video File Path:

Configuration

Multicast Group IP Address:

Multicast Group MAC Address:

Source IP Address:

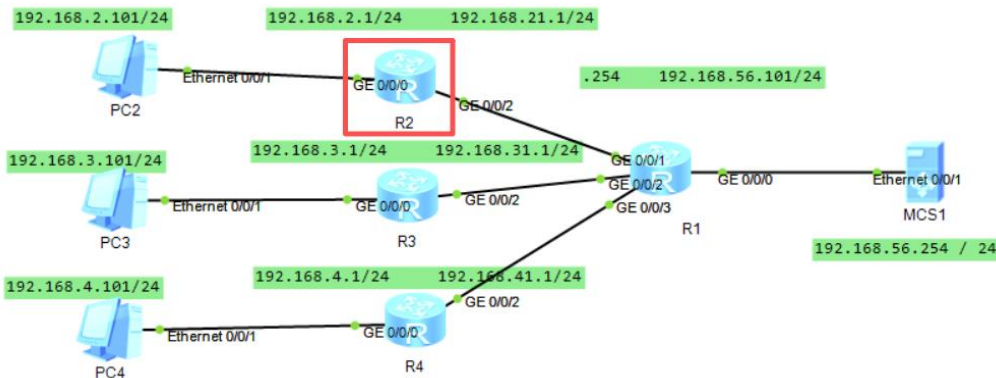
Source MAC Address:

Network Diagram: A topology showing four PCs (PC2, PC3, PC4) connected to three routers (R2, R3, R4). R2 is connected to PC2 and R3. R3 is connected to PC3 and R4. R4 is connected to PC4 and R1. R1 is connected to the MCS1 node. IP addresses are assigned to each device: PC2 (192.168.2.101/24), PC3 (192.168.3.1/24), PC4 (192.168.4.1/24), R2 (192.168.2.1/24), R3 (192.168.3.1/24), R4 (192.168.4.1/24), and MCS1 (192.168.56.254/24).

Demo1 (4)- Router(1)

Configuration on the **Router** which **near multicast group members**

- 1. Interface and Routing configuration based on unicast
- 2. Multicast configuration
 - 1) system level: **multicast routing-enable**
 - 2) interface level:
 - The interface near multicast group members
 - Enable IGMP: **igmp enable**
 - The interface near multicast source
 - Enable PIM DM: **pim dm**

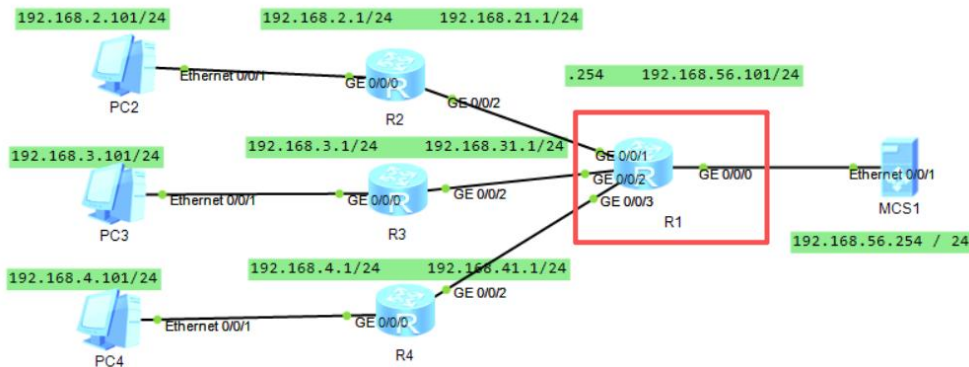


```
R2
R2
[Huawei]dis this
#
sysname Huawei
#
multicast routing-enable
#
return
[Huawei]int gi0/0/0
[Huawei-GigabitEthernet0/0/0]dis this
#
interface GigabitEthernet0/0/0
ip address 192.168.2.1 255.255.255.0
igmp enable
#
return
[Huawei-GigabitEthernet0/0/0]int gi0/0/2
[Huawei-GigabitEthernet0/0/2]dis this
#
interface GigabitEthernet0/0/2
ip address 192.168.21.1 255.255.255.0
pim dm
#
return
```


Demo1(5) - Router(2)

Configuration on the **Router** which **near multicast source**

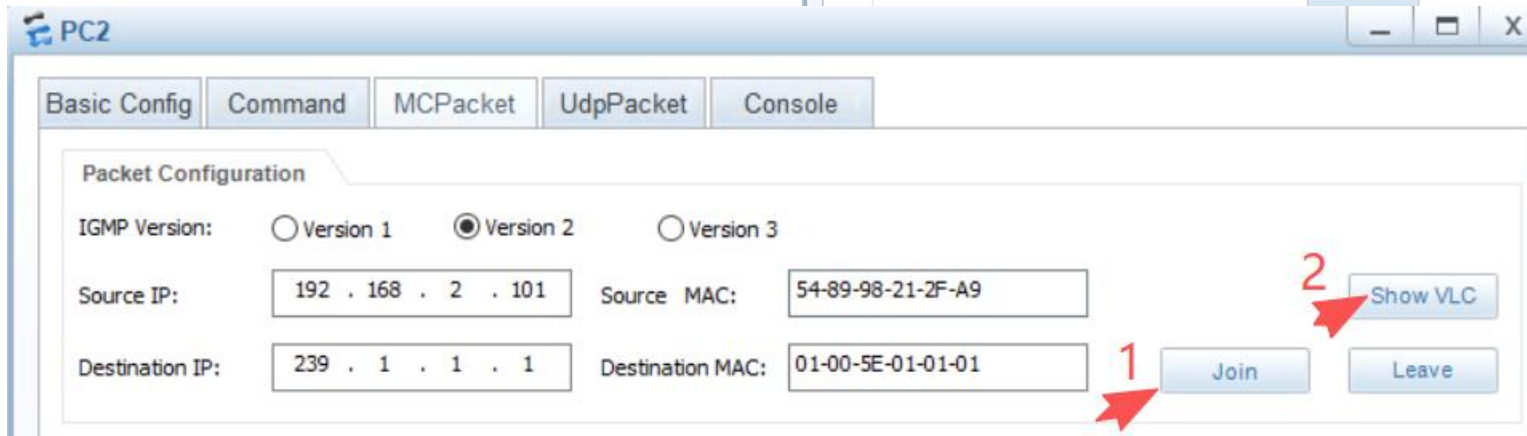
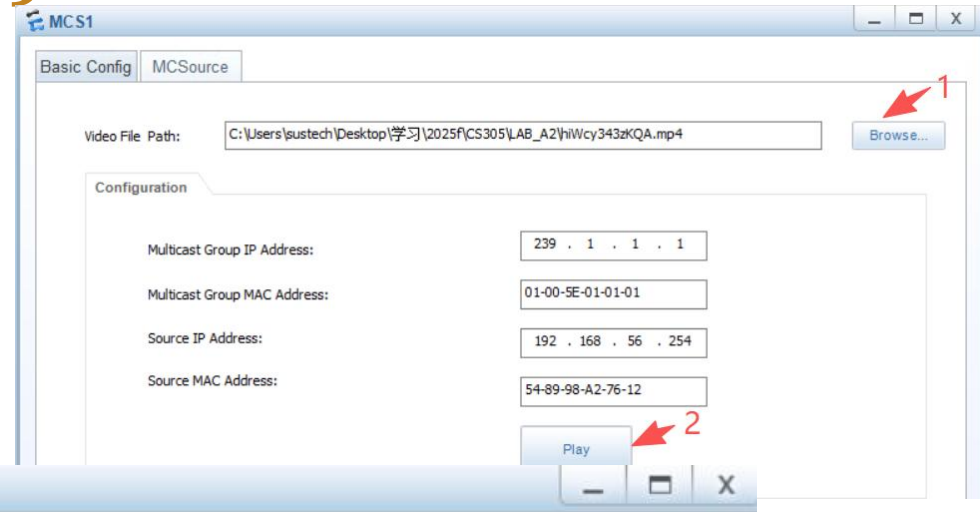
- 1. Interface and Routing configuration based on unicast
- 2. Multicast configuration
 - 1) system level: **multicast routing-enable**
 - 2) interface level:
 - ALL the interfaces
 - Enable IGMP: **pim dm**



```
R1
[Huawei]dis this
#
sysname Huawei
#
multicast routing-enable
#
return
[Huawei]int gi 0/0/1
[Huawei-GigabitEthernet0/0/1]dis this
#
interface GigabitEthernet0/0/1
ip address 192.168.21.254 255.255.255.0
pim dm
#
return
[Huawei-GigabitEthernet0/0/1]int gi0/0/0
[Huawei-GigabitEthernet0/0/0]dis this
#
interface GigabitEthernet0/0/0
ip address 192.168.56.101 255.255.255.0
pim dm
#
return
[Huawei-GigabitEthernet0/0/0]
```


Demo1 (6)- Test(1)

- **MSC**
 - Select an MP4 file(click “Browse”) and click “Play”
- **Multicast group member**
 - Join the multicast group(click “Join”) and click “Show VLC”

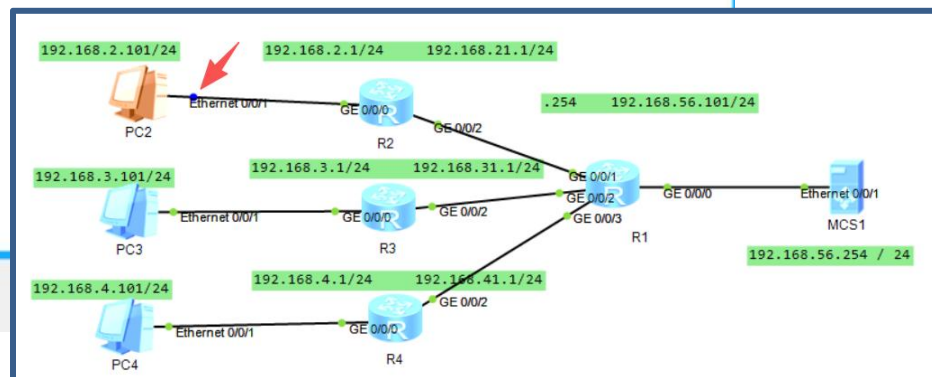


Demo1 (7)- Test(2)

Multicast packets captured on the interface Ethernet 0/0/1 of multicast group member (PC2)

No.	Time	Source	Protocol	Destination	Info
28638	198.0470...	192.168.56.254	UDP	239.1.1.1	60903 → 0 Len=1328
28639	198.0470...	192.168.56.254	UDP	239.1.1.1	60903 → 0 Len=1328

> Frame 28638: 1370 bytes on wire (10960 bits), 1370 bytes captured (10960 bits) on interface - id 0
Ethernet II, Src: HuaweiTe fd:64:76 (54:89:98:fd:64:76), Dst: IPv4mcast 01:01:01 (01:00:5e:01:01:01)
Internet Protocol Version 4, Src: 192.168.56.254, Dst: 239.1.1.1
0100 = Version: 4
.... 0101 = Header Length: 20 bytes (5)
> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 1356
Identification: 0x7f66 (32614)
> 000. = Flags: 0x0
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 126
Protocol: UDP (17)
Header Checksum: 0xce91 [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.56.254
Destination Address: 239.1.1.1
> User Datagram Protocol, Src Port: 60903, Dst Port: 0
> Data (1328 bytes)

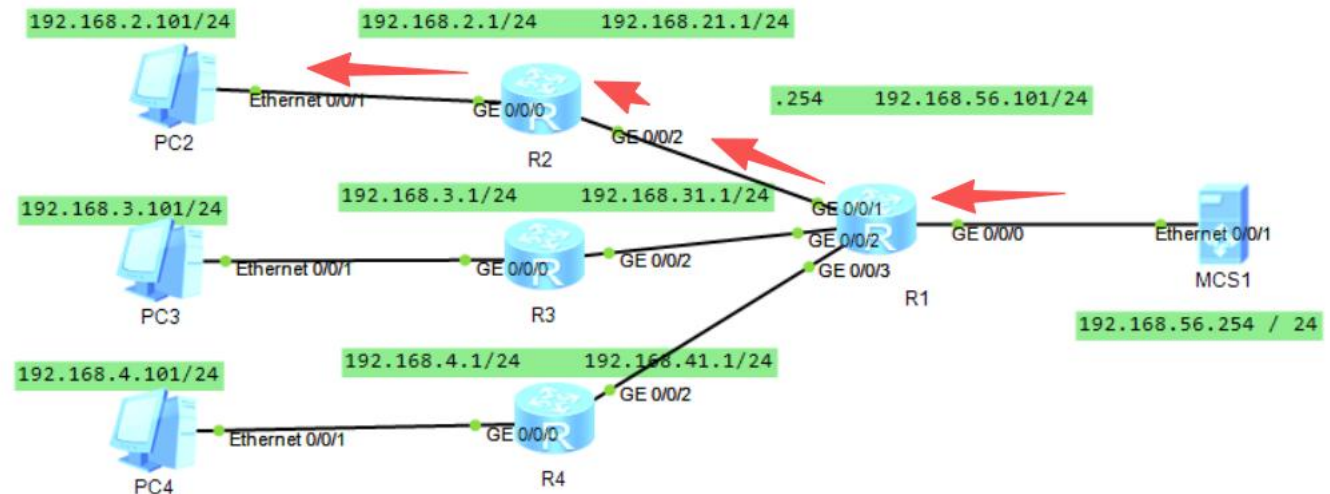


Demo1(8)- Test(3)

Command “**dis multicast forwarding-table**” is used in system-view on the router which process multicast forwarding. The **forwarding path of pkts**(from 192.168.56.254 to 239.1.1.1) on R1 is from its GE 0/0/0 to its GE 0/0/1, whild on R2 is from its GE 0/0/2 to its GE0/0/0.

This command can also display **forwarding statistics information about multicast packets**. Please **note that this statistics information has an aging time**. If there are no multicast messages after a period of time(less than a minute), the statistics information will be reset to 0.

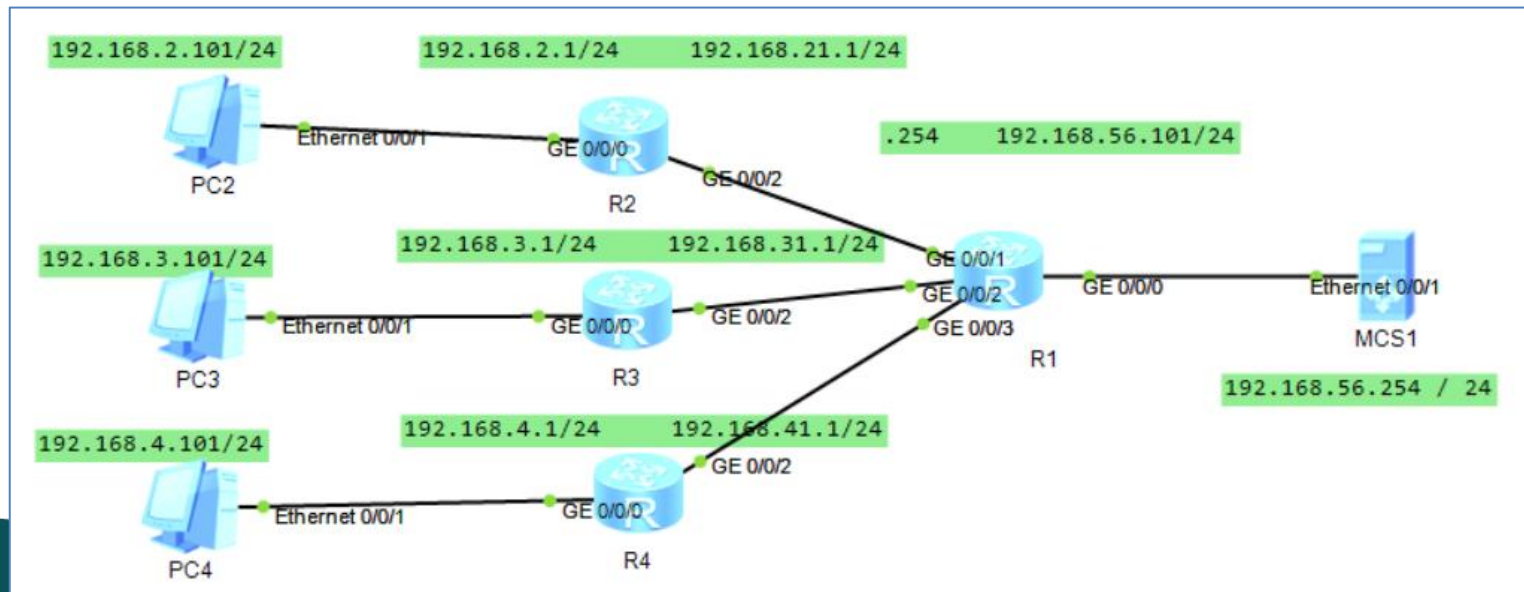
R1	R2
<pre>[Huawei]dis multicast forwarding-table Multicast Forwarding Table of VPN-Instance: publ Total 1 entry, 1 matched 00001. (192.168.56.254, 239.1.1.1) MID: 0, Flags: ACT Uptime: 00:01:03, Timeout in: 00:02:42 Incoming interface: GigabitEthernet0/0/0 List of 1 outgoing interfaces: 1: GigabitEthernet0/0/1 Matched 10654 packets(14446824 bytes), Wron Forwarded 10654 packets(14446824 bytes)</pre>	<pre>[Huawei]dis multicast forwarding-table Multicast Forwarding Table of VPN-Instance: public net Total 1 entry, 1 matched 00001. (192.168.56.254, 239.1.1.1) MID: 0, Flags: ACT Uptime: 00:01:11, Timeout in: 00:02:46 Incoming interface: GigabitEthernet0/0/2 List of 1 outgoing interfaces: 1: GigabitEthernet0/0/0 Matched 10654 packets(14446824 bytes), Wrong If 0 packets Forwarded 10654 packets(14446824 bytes)</pre>



Practise 10.1

1. Refer to Demo1 to build an IPv4 based multicast network (unicast and multicast addresses should be set by yourself, and they should be different from the addresses in Demo1). What's the IPv4 address of MCS, what's the multicast IPv4 address and the MAC address?
2. Play videos at the multicast source, and record the multicast forwarding information on the router corresponding to changes in multicast group members.

Multicast group	R1 incoming interface	R1 outgoing interface	R2 incoming interface	R2 outgoing interface	R3 incoming interface	R3 outgoing interface	R4 incoming interface	R4 outgoing interface	PC4 receive multicast pkt (Y/N)
PC2, PC3									
PC2, PC3, PC4									



Practise 10.2

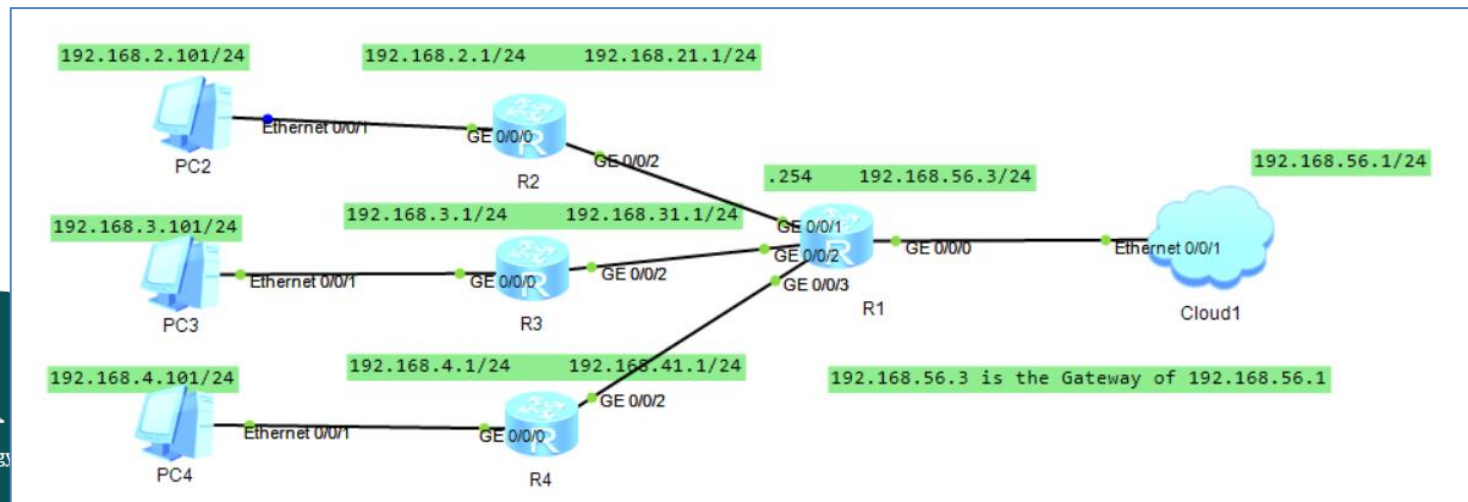
1. build an IPv4 based multicast network as follow topology, use a computer running eNSP as the multicast source, which communicates with the virtual network card (VirtualBox Host Only Network / Microsoft KM-TEST loopback adapter, its IPv4 address is 192.168.56.1) and cloud devices in the eNSP virtual network.

2. Run the “ping 192.168.2.101 -S 192.168.56.1” in the command line window to send uicast pkts to test determine whether the unicast network is connected. NOTES: if the unicast network is connected, ICMP reply message should be received.

3. Run the “ping 239.1.1.1 -S 192.168.56.1” in the command line window to send multicast(supose the multicast address is 239.1.1.1)

- ✓ capture packets, or run the “dis multicast forwarding-table ” to view the number of multicast pkts forwards on each device interface on the Router, and record it (the number of pkts) in the following table.
- ✓ Is there any ICMP reply message duirning the multicast test?
- ✓ Compare the difference between unicast and multicast.

Multicast group	Multicast source node 192.168.56.1 TX pkts	R1 GE0 RX pkts	R1 GE1 TX pkts	R1 GE2 TX pkts	R1 GE3 TX pkts	R2 GE0 TX pkts	R3 GE0 TX pkts	R4 GE0 TX pkts
PC2, PC3								
PC2, PC3, PC4								

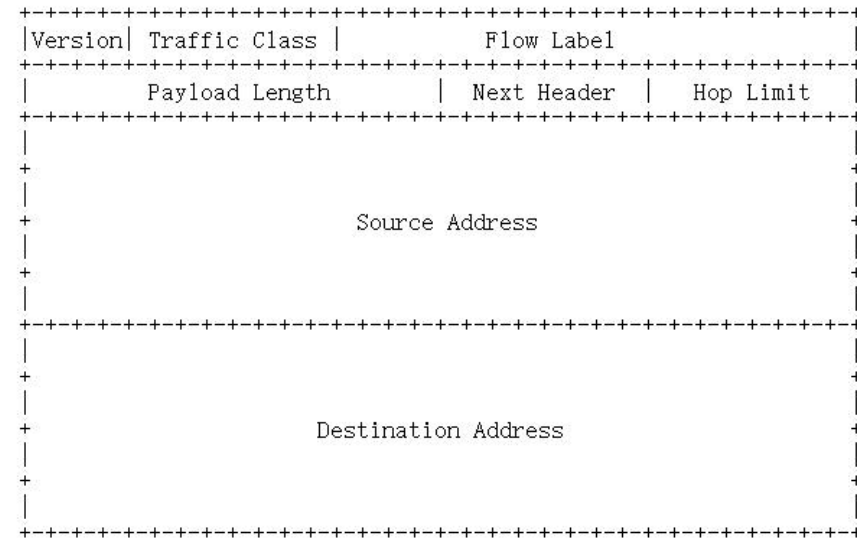


Part B. IPv6

- **IPv6** is a new version of the Internet Protocol, designed as the successor to IPv4. The changes from IPv4 to IPv6 fall primarily into the following categories:
 - **Expanded Addressing Capabilities:** IPv6 increases the IP address size from **32** bits to **128** bits, to support more levels of addressing hierarchy, a much greater number of addressable nodes, and simpler auto-configuration of addresses. The scalability of multicast routing is improved by adding a "scope" field to multicast addresses. And a new type of address called an "anycast address" is defined, used to send a packet to any one of a group of nodes.
 - **Header Format Simplification:** Some IPv4 header fields have been dropped or made optional, to reduce the common-case processing cost of packet handling and to limit the bandwidth cost of the IPv6 header.
 - **Improved Support for Extensions and Options:** Changes in the way IP header options are encoded allows for more efficient forwarding, less stringent limits on the length of options, and greater flexibility for introducing new options in the future.
 - **Flow Labeling Capability:** A new capability is added to enable the labeling of packets belonging to particular traffic "flows" for which the sender requests special handling, such as non-default quality of service or "real-time" service.
 - **Authentication and Privacy Capabilities:** Extensions to support authentication, data integrity, and (optional) data confidentiality are specified for IPv6.

IPv6 Header(1)

- **Version**
4-bit Internet Protocol version number = 6.
- **Traffic Class**
8-bit traffic class field.
- **Flow Label**
20-bit flow label.
- **Payload Length**
16-bit unsigned integer. **Length of the IPv6 payload,** i.e., the rest of the packet following this IPv6 header, in octets. (Note that any extension headers present are considered part of the payload, i.e., included in the length count.)
- **Next Header**
8-bit selector. Identifies the type of header immediately following the IPv6 header.
- **Hop Limit**
8-bit unsigned integer. Decremented by 1 by each node that forwards the packet. The packet is discarded if Hop Limit is decremented to zero.



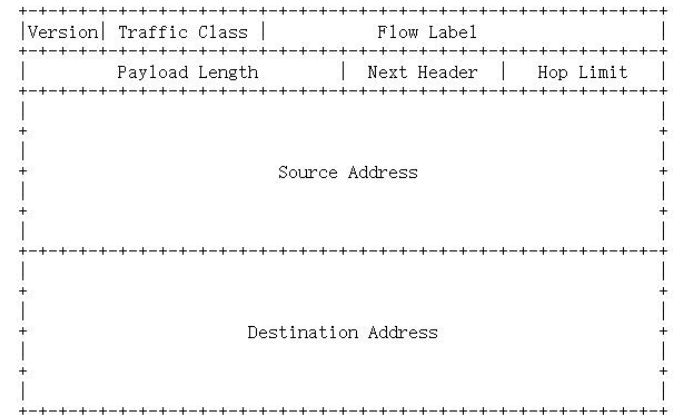
- **Source Address**
128-bit address of the originator of the packet
- **Destination Address**
128-bit address of the intended recipient of the packet. (possibly not the ultimate recipient, if a Routing header is present)

IPv6 Header(2)

```
icmpv6
No.      Time      Source      Destination  Protocol  Length  Info
1 0.000000  ::1        ::1          ICMPv6      84      Echo (ping) request id=0x0001, seq=
<
> Null/Loopback
v Internet Protocol Version 6, Src: ::1, Dst: ::1
  0110 .... = Version: 6
  > .... 0000 0000 .... = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)
  .... 0000 0000 0000 0000 0000 0000 = Flow Label: 0x00000
  Payload Length: 40
  Next Header: ICMPv6 (58)
  Hop Limit: 64
  Source: ::1
  Destination: ::1
v Internet Control Message Protocol v6
  Type: Echo (ping) request (128)
  Code: 0
  Checksum: 0xd4a6 [correct]
  [Checksum Status: Good]
  Identifier: 0x0001
  Sequence: 80
  [Response In: 2]
v Data (32 bytes)
  Data: 6162636465666768696a6b6c6d6e6f707172737475767761...
  [Length: 32]

0000  18 00 00 00 60 00 00 00  00 28 3a 40 00 00 00 00  .....`... .(:@...
0010  00 00 00 00 00 00 00 00  00 00 00 01 00 00 00 00  .....
0020  00 00 00 00 00 00 00 00  00 00 00 01 80 00 d4 a6  .....

Internet Protocol Version 6 (ipv6), 40 byte(s)  分组: 161
```



using 'ping -6 localhost'
to invoke an ICMPv6
transaction.

```
ICMPv6. type : 128
ICMPv6. code : 0
```

ICMPv6 Echo request

IPv6 Header(3)

In this situation, there is no extension header, the header of upper layer(UDP, TCP, ICMPv6) is the next header.

```
Internet Protocol Version 6, Src: fe80::e7be:bbb7:72b9:6ddc, Dst: ff02::fb
  0110 .... = Version: 6
  > .... 0000 0000 .... = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)
  .... 0011 0101 1101 1000 0001 = Flow Label: 0x35d81
  Payload Length: 47
  Next Header: UDP (17)
  Hop Limit: 1
  Source Address: fe80::e7be:bbb7:72b9:6ddc
  Destination Address: ff02::fb
  > User Datagram Protocol, Src Port: 5353, Dst Port: 5353
```

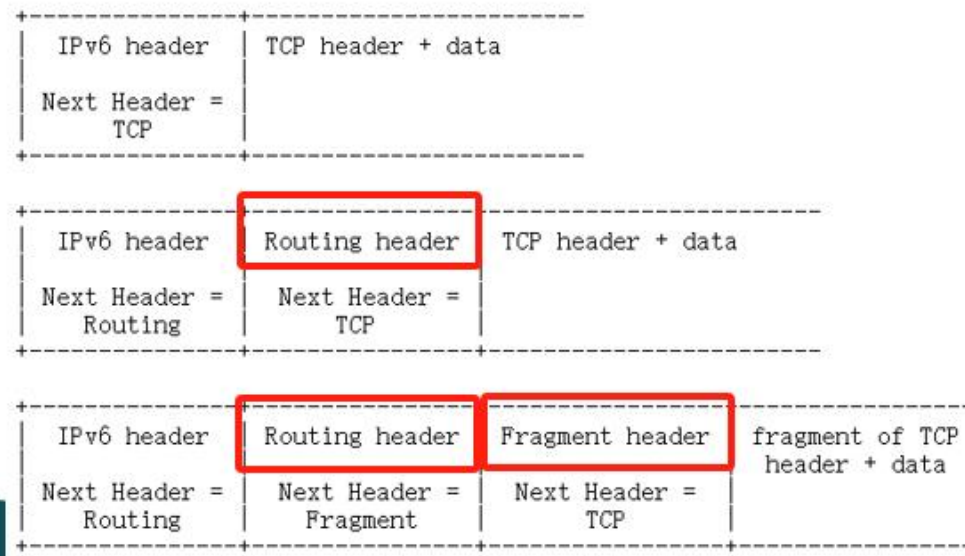


```
Internet Protocol Version 6, Src: 2001:da8:201d:1109::fcd7, Dst: 2600:140e:6:9b5::2c1a
  0110 .... = Version: 6
  > .... 0000 0000 .... = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)
  .... 1010 0011 0111 0111 1100 = Flow Label: 0xa377c
  Payload Length: 203
  Next Header: TCP (6)
  Hop Limit: 64
  Source Address: 2001:da8:201d:1109::fcd7
  Destination Address: 2600:140e:6:9b5::2c1a
  Transmission Control Protocol, Src Port: 64779, Dst Port: 443, Seq: 1, Ack: 1, Len: 183
  Transport Layer Security
```

```
Internet Protocol Version 6, Src: 2001:da8:201d:1109::fcd7, Dst: 2001:da8:201d::42:3
  0110 .... = Version: 6
  > .... 0000 0000 .... = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)
  .... 0000 0000 0000 0000 0000 = Flow Label: 0x00000
  Payload Length: 40
  Next Header: ICMPv6 (58)
  Hop Limit: 128
  Source Address: 2001:da8:201d:1109::fcd7
  Destination Address: 2001:da8:201d::42:3
  > Internet Control Message Protocol v6
```

IPv6 Extension Headers(1)

- In IPv6, **optional** internet-layer information is encoded in **separate headers that may be placed between the IPv6 header and the upper-layer header** in a packet. There is a small number of such extension headers, each one identified by a distinct Next Header value.
 - **Extension headers (except for the Hop-by-Hop Options header) are NOT processed, inserted, or deleted by any node along a packet's delivery path, until the packet reaches the node (or each of the set of nodes, in the case of multicast) identified in the Destination Address field of the IPv6 header.**
 - **an IPv6 packet may carry zero, one, or more extension headers, each identified by the Next Header field of the preceding header.**



IPv6 Address(1)

- Text Representation of Addresses
 - The preferred form is **x:x:x:x:x:x:x**, where the 'x's are the **hexadecimal** values of the eight **16-bit** pieces of the address
 - In order to make writing addresses containing zero bits easier a special syntax is available to compress the zeros. The use of “::” indicates **multiple groups of 16-bits of zeros**. The “::” can only appear once in an address.
- Address Type Representation
 - The address **0:0:0:0:0:0:0:0** is called the unspecified address.
 - The unicast address 0:0:0:0:0:0:0:1 is called the loopback address.
 - Link-Local Unicast Addresses are designed to be used for addressing on a single link for purposes such as auto-address configuration, neighbor discovery, or when no routers are present.

```
IPv6 地址 . . . . . : 2409:8a55:3050:f6b0:48cf:2c49:a3fe:6381
临时 IPv6 地址. . . . . : 2409:8a55:3050:f6b0:6cf9:d6fc:544:f4c7
本地链接 IPv6 地址. . . . . : fe80::48cf:2c49:a3fe:6381%17
```



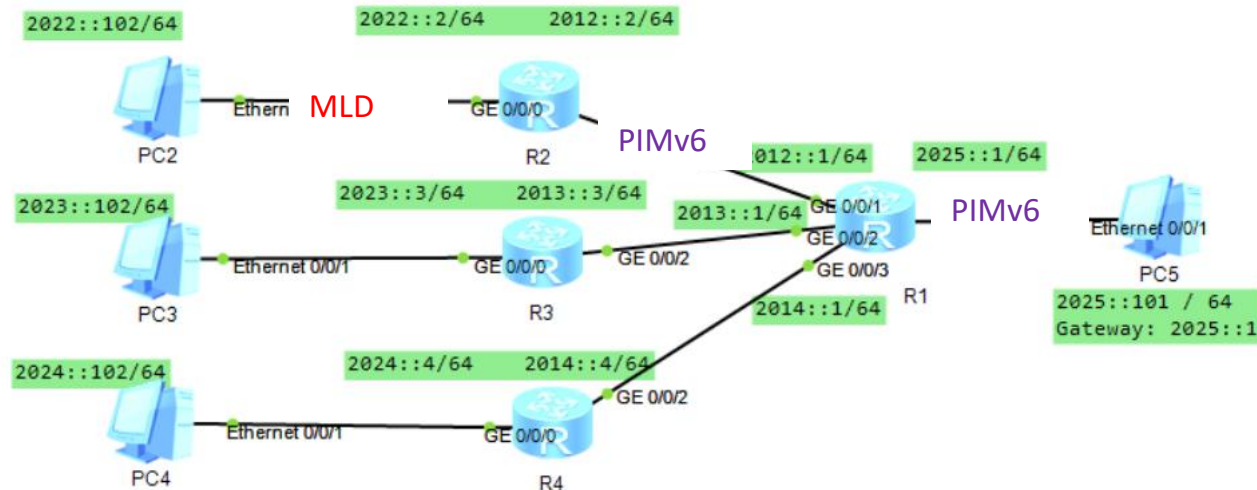
IPv6 Address(2)

- **Unicast Addresses:** Unicast addresses identify a single network interface for one-to-one communication. The main types include:
 - **Global Unicast Address:** Used for public internet routing, with a prefix range of **2000::/3** (2000:: to 3FFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF)
 - **Link-Local Address:** Used for local communication within a single network segment, with a fixed prefix of **FE80::/10** (typically **FE80::/64**)
 - **Unique Local Address (ULA):** Similar to IPv4 private addresses, used within private networks, with a prefix range of **FC00::/7** (commonly **FD00::/8** for local use)
- **Multicast Addresses:** Multicast addresses identify multiple interfaces for one-to-many communication. **All multicast addresses start with FF (1111 in binary)**
 - FF02::1: All nodes on the local link.
 - FF02::2: All routers on the local link.
 - .etc.

IPv6 Multicast

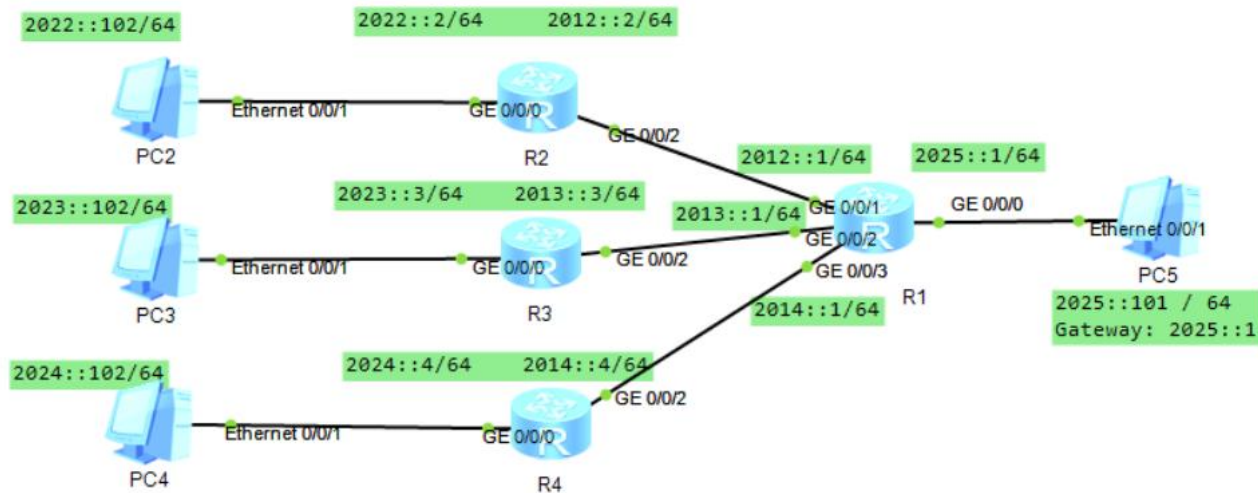
Relevant IPv6 Multicast Protocols

- **M**ulticast **L**istener **D**iscovery (**MLD**) protocol is responsible for managing IPv6 multicast group memberships between hosts and their directly connected multicast routers. It functions as the IPv6 counterpart to the Internet Group Management Protocol (IGMP) in IPv4, with identical protocol behavior but different message formats.
- The **P**rotocol **I**ndependent **M**ulticast (**PIM**) in **IPv6** is a routing protocol used to establish and maintain multicast routing trees for efficient data transmission in multicast networks. It operates independently of the unicast routing protocol and supports two main modes: Sparse Mode (PIM-SM) and Dense Mode (PIM-DM)



Demo2 (1) - topology

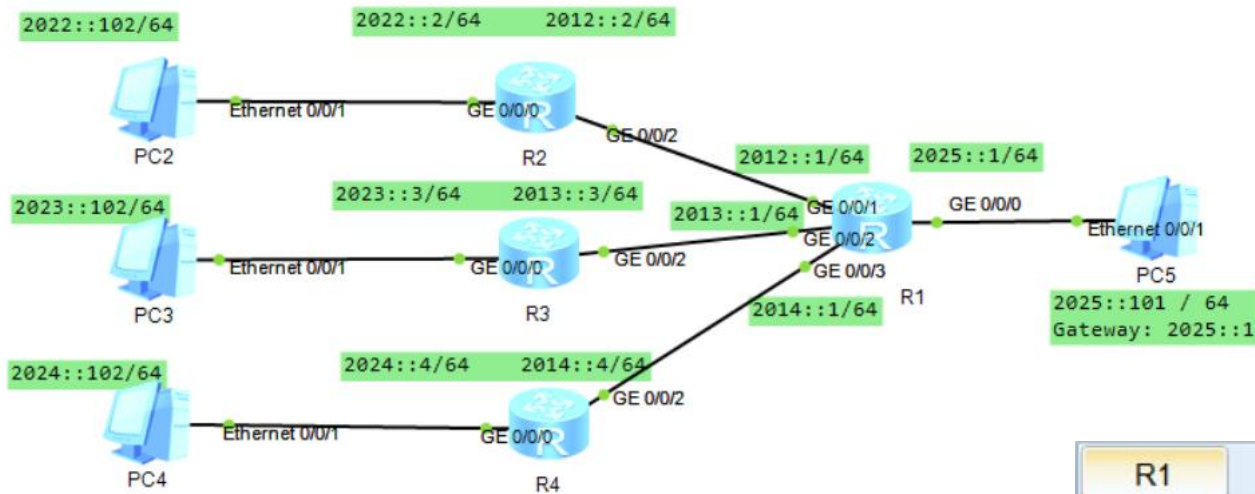
- In **IPv6** network topology:
 - Using **PC5** as multicast source node
 - Using interface **GE0/0/0** of **R2,R3,R4** as potential multicast group member nodes
 - Routers **R1, R2, R3, and R4** are routers between multicast source nodes and multicast group member nodes, **generating multicast related routes through PIMv6-DM protocol interaction**
 - **R2, R3, and R4** are routers near multicast group member nodes, **sensing changes in multicast groups through MLD protocol interaction.**



TIPS: Complete the network connectivity configuration based on unicast first, and then proceed with multicast configuration.

Demo2 (2) - IPv6 configuration(basic)

- **Basic configuration of IPv6 on the Router:**
 - **system view**
 - command on enable IPv6: **ipv6**
 - command on add static route item: **ipv6 route-static [destination network] [IPv6 address of next-hop]**
 - **interface view**
 - command on enable IPv6: **ipv6 enable**
 - command on configure IPv6 address: **ipv6 addr [IPv6 address] [IPv6 prefix length <1-128>]**



```
R1
[Huawei]dis this
#
sysname Huawei
#
ipv6
#
multicast routing-enable
multicast ipv6 routing-enable
#
ipv6 route-static 2022:: 64 2012::2
ipv6 route-static 2023:: 64 2013::3
ipv6 route-static 2024:: 64 2014::4
#
```

```
R1
[Huawei-GigabitEthernet0/0/0]dis this
#
interface GigabitEthernet0/0/0
ipv6 enable
ip address 192.168.56.3 255.255.255.0
ipv6 address 2025::1/64
pim dm
pim ipv6 dm
#
return
[Huawei-GigabitEthernet0/0/0]
```

IPv6 Configuration

☒ Static ☐ DHCPv6

IPv6 Address: 2022::102

Prefix Length: 64

IPv6 Gateway: 2022::2

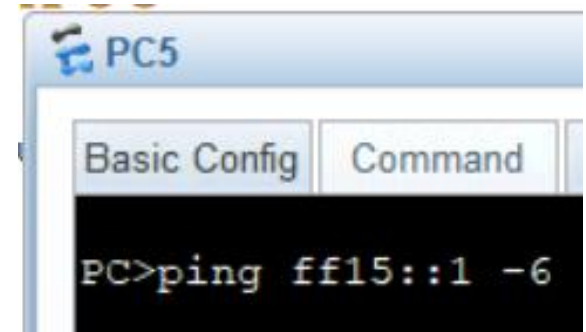
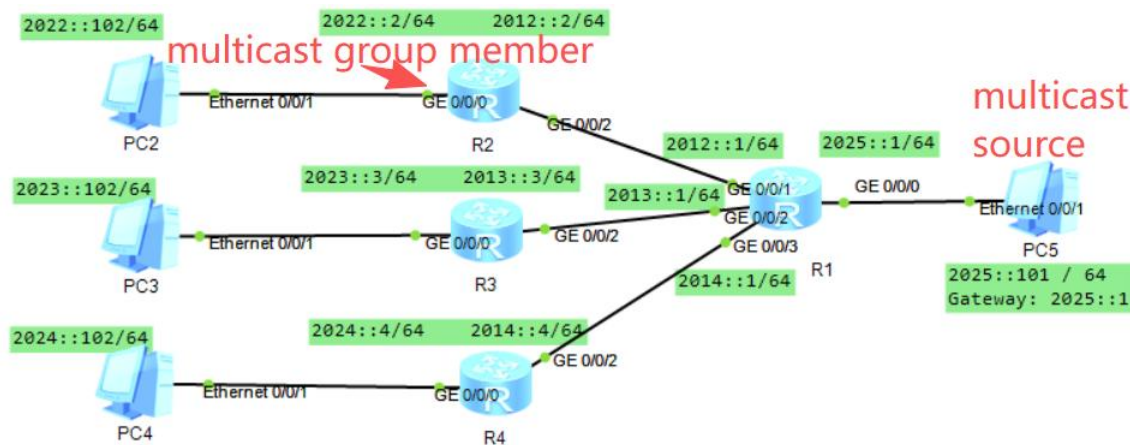
Demo 2(3) - multicast source

PC5 work as **multicast source node** in this IPv6 Multicast demo. While choose “FF15::1” as the multicast IP address, it’s MAC address is auto-generated as “33:33::1”.

Uing cmd “**ping [IPv6 multicast address] -6**” to send IPv6 multicast pkts.

MAC Address(48bits) Mapping:

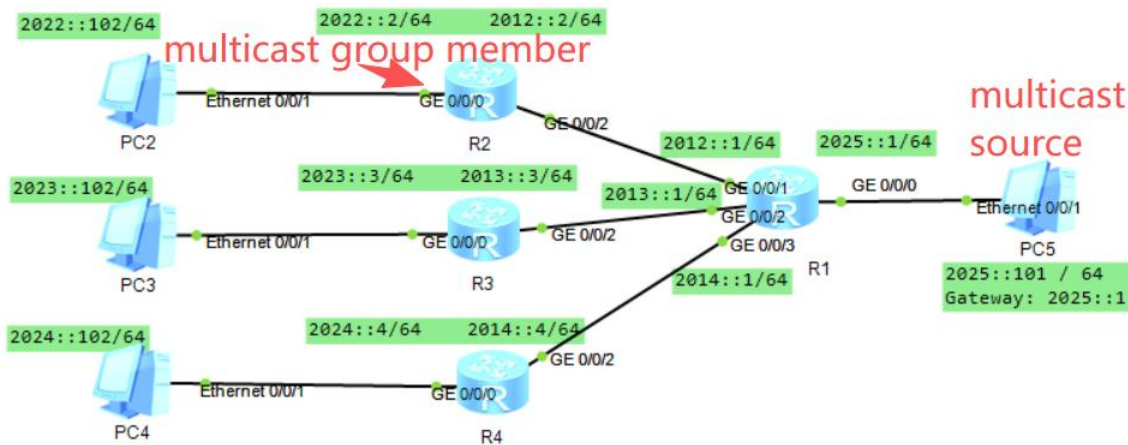
- **Prefix:** The MAC address always starts with 33-33 (hexadecimal).
- **Mapping:** The last 4 bytes (32 bits) of the IPv6 multicast address are directly mapped to the remaining 4 bytes of the MAC address.



icmpv6					
No.	Time	Source	Protocol	Destination	Info
2	9.688000	2025::101	ICMPv6	ff15::1	Echo (ping) request id=0x4619, seq=1,
<					
> Frame 2: 94 bytes on wire (752 bits), 94 bytes captured (752 bits) on interface -, id 0					
> Ethernet II, Src: HuaweiTe_fd:64:76 (54:89:98:fd:64:76), Dst: IPv6mcast_01 (33:33:00:00:00:01)					
> Internet Protocol Version 6, Src: 2025::101, Dst: ff15::1					
> Internet Control Message Protocol v6					

Demo 2(4) - multicast group member

In this IPv6 Multicast demo, the **R2,R3,R4's interface near to PC2,PC3,PC4** work as **multicast group member**. While choose “**FF15::1**” as the multicast IP address, it's MAC address is auto-generated as “**33:33::1**”.



TIPS: Since PCs in eNSP do not have IPv6 multicast functionality, this topology uses the interface near multicast members (assuming PC2, PC3, PC4 are potential multicast members) in the router to simulate multicast members.

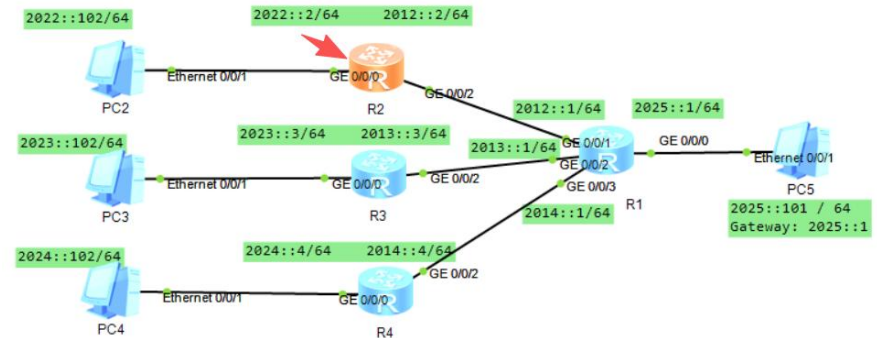
The configuration on the interface: “**mld static group FF15:: 1**” indicates **joining** the multicast group “FF15:: 1”, and “**undo mld static group FF15:: 1**” indicates **leaving** the multicast group “FF15:: 1”.

```
R2
[Huawei-GigabitEthernet0/0/0]dis this
#
interface GigabitEthernet0/0/0
ipv6 enable
ip address 192.168.2.1 255.255.255.0
ipv6 address 2022::2/64
igmp enable
igmp static-group 239.1.1.1
mld enable
mld static-group FF15::1
#
return
```

Demo2 (5)- Router(1)

Configuration on the **Router** which **near multicast group members**:

- 1. Interface and Routing configuration based on unicast
- 2. Multicast configuration
 - 1) system level: **multicast ipv6 routing-enable**
 - 2) interface level:
- The interface near multicast group members
 - Enable MLD: **mld enable**
- The interface near multicast source
 - Enable PIM v6 DM: **pim ipv6 dm**



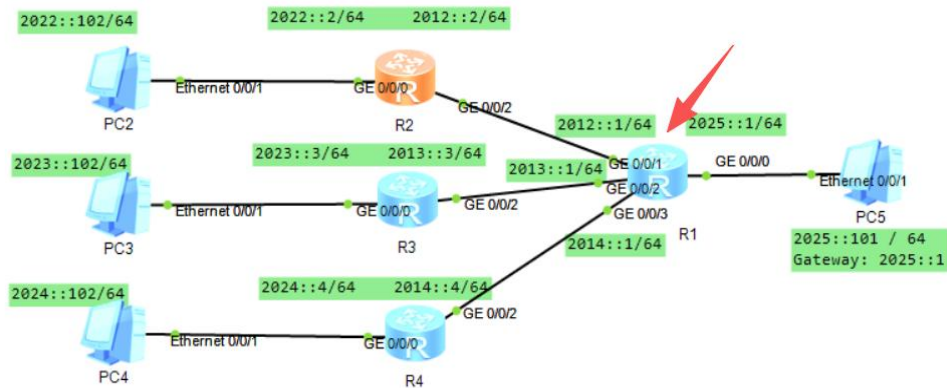
```
R2
[Huawei]dis this
#
sysname Huawei
#
ipv6
#
multicast routing-enable
multicast ipv6 routing-enable
#
ipv6 route-static 2025:: 64 2012::1
#
return
[Huawei]
```

```
R2
[Huawei]int gi 0/0/0
[Huawei-GigabitEthernet0/0/0]dis this
#
interface GigabitEthernet0/0/0
ipv6 enable
ip address 192.168.2.1 255.255.255.0
ipv6 address 2022::2/64
igmp enable
igmp static-group 239.1.1.1
mld enable
mld static-group FF15::1
#
return
[Huawei-GigabitEthernet0/0/0]int gi 0/0/2
[Huawei-GigabitEthernet0/0/2]dis this
#
interface GigabitEthernet0/0/2
ipv6 enable
ip address 192.168.21.1 255.255.255.0
ipv6 address 2012::2/64
pim dm
pim ipv6 dm
#
return
[Huawei-GigabitEthernet0/0/2]
```

Demo2(6) - Router(2)

Configuration on the **Router** which **near multicast source**:

- 1. Interface and Routing configuration based on unicast
- 2. Multicast configuration
 - 1) system level: **multicast ipv6 routing-enable**
 - 2) interface level(all the interface):
 - Enable PIM v6 DM: **pim ipv6 dm**



```
R1
[Huawei]dis this
#
sysname Huawei
#
ipv6
#
multicast routing-enable
multicast ipv6 routing-enable
#
ipv6 route-static 2022:: 64 2012::2
ipv6 route-static 2023:: 64 2013::3
ipv6 route-static 2024:: 64 2014::4
#
return
[Huawei]
```

```
R1
[Huawei]int gi 0/0/0
[Huawei-GigabitEthernet0/0/0]dis this
#
interface GigabitEthernet0/0/0
ipv6 enable
ip address 192.168.56.3 255.255.255.0
ipv6 address 2025::1/64
pim dm
pim ipv6 dm
#
return
[Huawei-GigabitEthernet0/0/0]int gi0/0/1
[Huawei-GigabitEthernet0/0/1]dis this
#
interface GigabitEthernet0/0/1
ipv6 enable
ip address 192.168.21.254 255.255.255.0
ipv6 address 2012::1/64
pim dm
pim ipv6 dm
#
return
[Huawei-GigabitEthernet0/0/1]
```

Demo2 (7)- Test(1)

IPv6 Multicast packets captured on the interface GE 0/0/0 of multicast group member (ff15::1)

icmpv6					
No.	Time	Source	Protocol	Destination	Info
4	25.625000	2025::101	ICMPv6	ff15::1	Echo (ping) request id=0x4aed, seq=1, hop
5	27.641000	2025::101	ICMPv6	ff15::1	Echo (ping) request id=0x4aee, seq=2, hop
6	29.672000	2025::101	ICMPv6	ff15::1	Echo (ping) request id=0x4aef, seq=3, hop
8	31.688000	2025::101	ICMPv6	ff15::1	Echo (ping) request id=0x4af0, seq=4, hop
9	33.704000	2025::101	ICMPv6	ff15::1	Echo (ping) request id=0x4af1, seq=5, hop

> Frame 4: 94 bytes on wire (752 bits), 94 bytes captured (752 bits) on interface -, id 0

✓ Ethernet II, Src: HuaweiTe_fd:64:76 (54:89:98:fd:64:76), Dst: IPv6mcast_01 (33:33:00:00:00:01)

> Destination: IPv6mcast_01 (33:33:00:00:00:01)

> Source: HuaweiTe_fd:64:76 (54:89:98:fd:64:76)

Type: IPv6 (0x86dd)

✓ Internet Protocol Version 6, Src: 2025::101, Dst: ff15::1

0110 = Version: 6

> 0000 0000 = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)

.... 0000 0000 0000 0000 0000 = Flow Label: 0x000000

Payload Length: 40

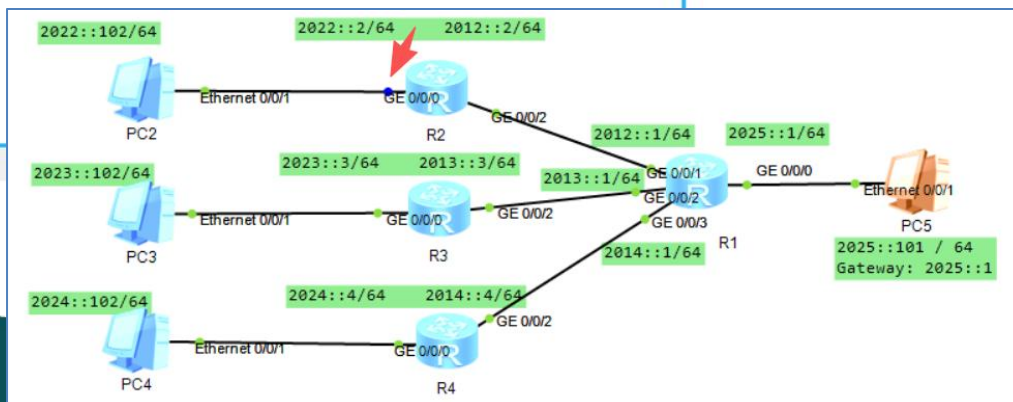
Next Header: ICMPv6 (58)

Hop Limit: 253

Source Address: 2025::101

Destination Address: ff15::1

> Internet Control Message Protocol v6



Demo2(8)- Test(2)

Command “**dis multicast ipv6 forwarding-table**” is used in system-view on the router which process IPv6 multicast forwarding. The **forwarding path of pkts**(from 2025::101 to FF15::1) on R1 is from its GE 0/0/0 to its GE 0/0/1, while on R2 is from its GE 0/0/2 to its GE0/0/0.

This command can also display **forwarding statistics information about multicast packets**. Please **note that this statistics information has an aging time**. If there are no multicast messages after a period of time(less than a minute), the statistics information will be reset to 0.

```
[Huawei]dis multicast ipv6 forwarding-table
IPv6 Multicast Forwarding Table
Total 1 entry, 1 matched

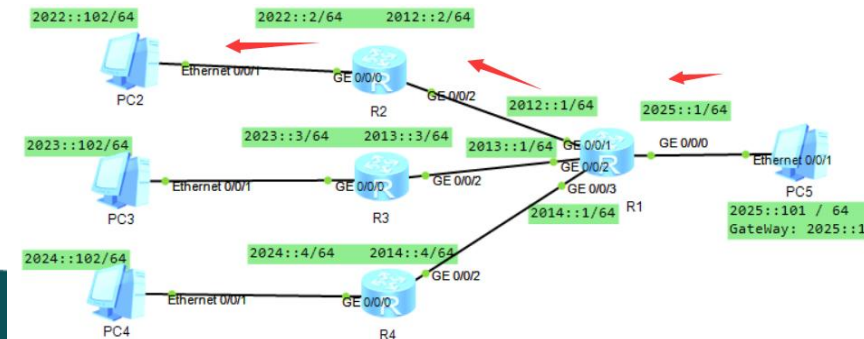
00001. (2025::101, FF15::1)
  MID: 0, Flags: ACT
  Uptime: 00:00:17, Timeout in: 00:03:26
  Incoming interface: GigabitEthernet0/0/2
  List of 1 outgoing interfaces:
    1: GigabitEthernet0/0/0
  Matched 5 packets(51200 bytes), Wrong If 0 packets
  Forwarded 5 packets(51200 bytes)

[Huawei]
```

```
[Huawei]dis multicast ipv6 forwarding-table
IPv6 Multicast Forwarding Table
Total 1 entry, 1 matched

00001. (2025::101, FF15::1)
  MID: 0, Flags: ACT
  Uptime: 00:00:13, Timeout in: 00:03:18
  Incoming interface: GigabitEthernet0/0/0
  List of 3 outgoing interfaces:
    1: GigabitEthernet0/0/1
    2: GigabitEthernet0/0/2
    3: GigabitEthernet0/0/3
  Matched 5 packets(51200 bytes), Wrong If 0 packets
  Forwarded 5 packets(51200 bytes)

[Huawei]
```



```
PC>ping ff15::1 -6

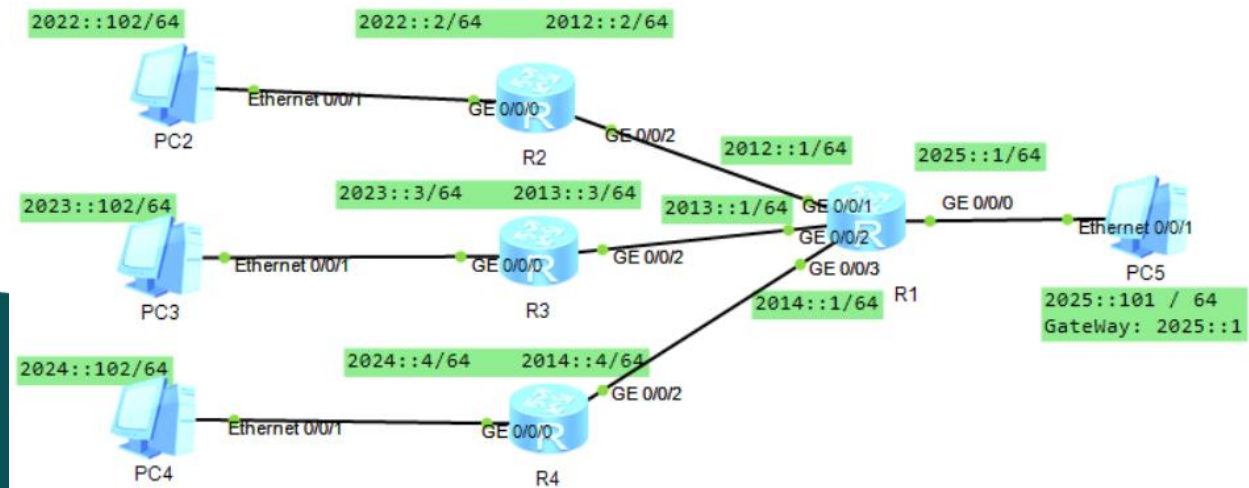
Ping ff15::1: 32 data bytes, Press Ctrl_C to break
Request timeout!
Request timeout!
Request timeout!
Request timeout!
Request timeout!

--- ff15::1 ping statistics ---
 5 packet(s) transmitted
 0 packet(s) received
100.00% packet loss
```


Practise 10.3

1. Build an IPv6 based multicast network as follow topology, use PC5 as the multicast source , use GE 0/0/0 of R2, R3 and R4 as potential multicast group member.
2. Run the “ping 2022::102 -6”, “ping 2023::102 -6”, “ping 2024::102 -6” in the command line window of PC5 to send uicast pkts to test whether the unicast network is connected. NOTES: if the unicast network is connected, ICMPV6 reply message should be received.
3. Run the “ping ff15::1” in the command line window of PC5 to send multicast(supose the multicast address is ff15::1)
 - ✓ capture packets, or run the “dis multicast ipv6 forwarding-table ” to view the number of multicast pkts forwards on each device interface on the Router, and record it (the number of pkts) in the following table.
 - ✓ Is there any ICMP reply message duirning the multicast test?
 - ✓ Compare the difference between unicast and multicast in IPv6.

Multicast group	PC5 TX pkts	R1 GE0 RX pkts	R1 GE1 TX pkts	R1 GE2 TX pkts	R1 GE3 TX pkts	R2 GE0 TX pkts	R3 GE0 TX pkts	R4 GE0 TX pkts
PC2(GE0/0/0 of R2), PC3(GE0/0/0 of R3)								
PC2(GE0/0/0 of R2), PC3(GE0/0/0 of R3), PC4(GE0/0/0 of R4)								



TIPS on VLC

VLC is a free and open-source cross platform multimedia player and framework that can play most multimedia files, as well as DVDs, audio CDs, VCDs, and various streaming protocols.

eNSP use VLC to play the video on both multicast source and group member.

The VLC installation package could be get from : <https://www.videolan.org/vlc/>

After install the VLC, set its path in eNSP's "Tools" setting item, then click on "OK".

